Indian (Hindu) Time-Keeping: Weekdays, Tithis, Nakshatras and the Panchang

What you will be able to do

1. Explain how the **order of the weekdays** comes from the speeds of the "planets" across the sky.
2. Define a **tithi** and calculate it from the Moon–Sun angle.
3. Name the 15 tithis, the 12 lunar months and the 27 nakshatras, and link each month to its nakshatra.
4. Explain why an **Adhik Maas** (extra month) is needed and how often it occurs.
5. Convert between Indian time units (ghati, pala, muhurta, prahar) and hours and minutes.
6. Explain why festivals such as Diwali and Holi move around the Gregorian calendar.

1. The Panchang: "five limbs" of the day

A traditional Indian almanac, the **Panchang** (*pancha* = five, *anga* = limb), describes each day by five astronomical quantities:

Limb	What it measures	Based on
Vara	weekday	the hora (planetary hour) system
Tithi	lunar day	the Moon–Sun angle, in steps of 12°
Nakshatra	lunar mansion	the Moon's position among the stars, in steps of 13°20′
Yoga	a combined angle	(Sun's longitude + Moon's longitude), in steps of 13°20′
Karana	half a tithi	every 6° of Moon–Sun angle

Computing a Panchang needs real astronomy: the positions of the Sun and the Moon. For over 1,500 years Indian astronomers — Aryabhata, Brahmagupta, Bhaskara II and many others — refined these calculations.

A note on science

Historically, *Jyotisha* covered both astronomy (calculating positions, eclipses and calendars) and astrology (predictions about human lives). In this course we study only the **astronomy**, which is testable and precise. Astrological predictions have not been supported by scientific tests.

2. Vara: why the weekdays are in THIS order

Step 1 — order the seven "planets" by speed

The ancients knew seven bodies that move against the stars (*grahas*) and ranked them from **slowest** (assumed farthest) to **fastest** (nearest):

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P.T.O.

Rank	Body	Sanskrit	Time to circle the zodiac
1 (slowest)	Saturn	Shani	~29.5 years
2	Jupiter	Guru / Brihaspati	~11.9 years
3	Mars	Mangala	~1.9 years
4	Sun	Ravi / Surya	1 year
5	Venus	Shukra	(stays near the Sun)
6	Mercury	Budha	(stays near the Sun)
7 (fastest)	Moon	Soma / Chandra	~27.3 days

Venus and Mercury never stray far from the Sun, so on average they too take one year; the ancients placed them between the Sun and the Moon. This sequence is called the **Chaldean order**; the same order appears in the *Surya Siddhanta*.

Step 2 — each hour (hora) is ruled by the next planet

Divide each day into **24 horas**. The first hora belongs to the ruler of the day, and each following hora goes to the next planet down the list, cycling back to Saturn after the Moon.

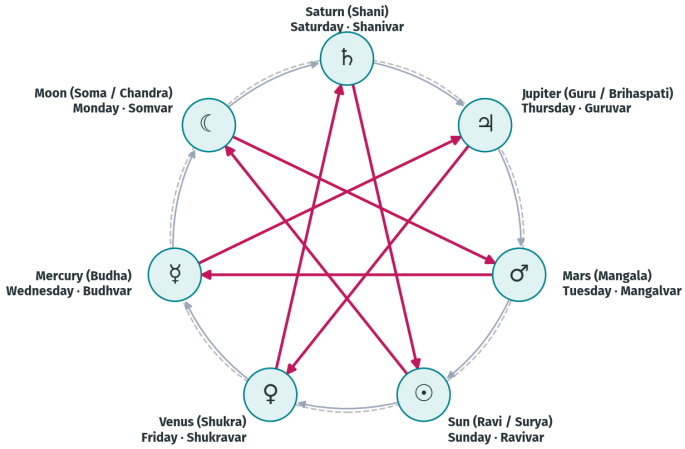
Step 3 — the ruler of the FIRST hora names the day

$$24 = 3 \times 7 + 3$$

After 24 horas we have gone round the list 3 times and moved **3 more places**. So the first hora of the next day belongs to the planet **3 places further down** the list:

Sun → Moon → Mars → Mercury → Jupiter → Venus → Saturn → Sun …

The 7-pointed star (heptagram) that names our weekdays



Grey circle: planets ordered from SLOWEST to FASTEST across the sky (Saturn → Moon).
Each hour (hora) of the day is ruled by the next planet in this order. 24 hours later the ruler has moved 24 = 3×7 + 3 places, i.e. 3 steps on. Follow the RED star: Sun → Moon → Mars → Mercury → Jupiter → Venus → Saturn → Sun … exactly the order of the weekdays!

Planets around a circle in speed order; jumping 3 places each time draws a 7-pointed star that gives the weekday order.

Day	Planet	Hindi/Sanskrit	Tamil	French	English origin
Sunday	Sun	**Ravivar**	Nyayiru	dimanche	Sun's day

(continued)

Day	Planet	Hindi/Sanskrit	Tamil	French	English origin
Monday	Moon	**Somvar**	Thingal	lundi	Moon's day
Tuesday	Mars	**Mangalvar**	Sevvai	mardi	Tiw's day (Norse god of war = Mars)
Wednesday	Mercury	**Budhvar**	Budhan	mercredi	Woden's day (= Mercury)
Thursday	Jupiter	**Guruvar**	Viyazhan	jeudi	Thor's day (= Jupiter)
Friday	Venus	**Shukravar**	Velli	vendredi	Frigg's day (= Venus)
Saturday	Saturn	**Shanivar**	Sani	samedi	Saturn's day

The same planetary week is used in India, Europe and much of Asia — a shared astronomical heritage about 2,000 years old. The Sanskrit word *hora* is related to the English *hour*.

3. Tithi: the lunar day

A **tithi** is the time taken by the Moon to move **12° ahead** of the Sun.

$$\frac{360^\circ}{12^\circ} = 30 \text{ tithis per lunar month}, \qquad \frac{29.53 \text{ d}}{30} \approx 23 \text{ h } 37 \text{ min per tithi (average)}.$$

Because the Moon's speed varies, a tithi can last anywhere from about **20 to 27 hours**. The tithi running **at sunrise** is the tithi of that day. Sometimes a tithi begins and ends between two sunrises and is "skipped" (*kshaya tithi*); sometimes one tithi covers two sunrises and is repeated (*adhika tithi*).

The two fortnights (paksha)

Shukla Paksha (waxing, bright)		Krishna Paksha (waning, dark)	
1	Pratipada	16	Pratipada
2	Dwitiya	17	Dwitiya
3	Tritiya	18	Tritiya
4	**Chaturthi**	19	**Chaturthi**
5	Panchami	20	Panchami
6	Shashthi	21	Shashthi
7	Saptami	22	Saptami
8	**Ashtami** (half Moon)	23	**Ashtami** (half Moon)
9	Navami	24	Navami
10	Dashami	25	Dashami
11	**Ekadashi**	26	**Ekadashi**
12	Dwadashi	27	Dwadashi
13	Trayodashi	28	Trayodashi
14	Chaturdashi	29	Chaturdashi
15	**Purnima** (Full Moon)	30	**Amavasya** (New Moon)

The names are Sanskrit ordinal numbers: *prathama* (first), *dwitiya* (second), *chaturthi* (fourth), *ekadashi* (eleventh)…

Calculating the tithi

If the Moon is E degrees ahead of the Sun (the elongation), then

$$\text{tithi number} = \left\lceil \frac{E}{12^\circ} \right\rceil$$

($\lceil\ \rceil$ means "round *up*"). Numbers 1–15 are Shukla; for 16–30, subtract 15 and the tithi is Krishna.

Worked examples

1. $E = 100^\circ$: $100/12 = 8.33 \Rightarrow$ 9th tithi = **Shukla Navami**.
2. $E = 250^\circ$: $250/12 = 20.8 \Rightarrow$ 21st tithi; $21 - 15 = 6 \Rightarrow$ **Krishna Shashthi**.
3. On Shukla Ekadashi (11th tithi), E lies between $10 \times 12^\circ = 120^\circ$ and $11 \times 12^\circ = 132^\circ$.

Festivals fixed by tithi (examples)

Festival	Tithi	Festival	Tithi
Gudi Padwa / Ugadi	Chaitra Shukla Pratipada	Navratri → Dussehra	Ashvin Shukla 1 → 10
Rama Navami	Chaitra Shukla Navami	Sharad Purnima	Ashvin Purnima
Akshaya Tritiya	Vaishakha Shukla Tritiya	**Diwali** (Lakshmi Puja)	**Amavasya** of Ashvin/Kartika
Buddha Purnima	Vaishakha Purnima	Chhath	Kartika Shukla Shashthi
Guru Purnima	Ashadha Purnima	Maha Shivaratri	Krishna Chaturdashi (Magha/Phalguna)
Raksha Bandhan	Shravana Purnima	**Holi**	**Phalguna Purnima**
Ganesh Chaturthi	Bhadrapada Shukla Chaturthi	Ekadashi fasts	every Shukla & Krishna Ekadashi

Because these festivals follow the Moon, they shift by up to about three weeks from year to year in the Gregorian calendar, whereas solar festivals such as Makar Sankranti (Lecture 1.2) hardly move.

4. The lunar months (masa) and how they got their names

Each lunar month is named after the **nakshatra near which the Full Moon occurs** in that month:

#	Month	Full Moon near	Approx.	#	Month	Full Moon near	Approx.
1	**Chaitra**	Chitra (Spica)	Mar–Apr	7	**Ashvin**	Ashvini	Sep–Oct
2	**Vaishakha**	Vishakha	Apr–May	8	**Kartika**	Krittika (Pleiades)	Oct–Nov
3	**Jyeshtha**	Jyeshtha (Antares)	May–Jun	9	**Margashirsha**	Mrigashira	Nov–Dec
4	**Ashadha**	Purva/Uttara Ashadha	Jun–Jul	10	**Pausha**	Pushya	Dec–Jan
5	**Shravana**	Shravana (Altair)	Jul–Aug	11	**Magha**	Magha (Regulus)	Jan–Feb
6	**Bhadrapada**	Purva/Uttara Bhadrapada	Aug–Sep	12	**Phalguna**	Purva/Uttara Phalguni	Feb–Mar

So the calendar itself is a star map: the Kartika Full Moon really does shine near the Pleiades every year.

Amanta and Purnimanta

- **Amanta** ("ending at Amavasya"): the month runs New Moon to New Moon — Maharashtra, Gujarat, Karnataka, Andhra Pradesh, Telangana, Goa.
- **Purnimanta** ("ending at Purnima"): Full Moon to Full Moon — most of north India (UP, Bihar, MP, Rajasthan, Punjab, Himachal...).

The Shukla paksha has the same month name in both systems, but the Krishna paksha takes the *next* month's name in Purnimanta. Example: Janmashtami is "Shravana Krishna Ashtami" (Amanta) but "Bhadrapada Krishna Ashtami" (Purnimanta) — the same night!

5. Adhik Maas: keeping the months in step with the seasons

$$12 \times 29.53 = 354.4 \text{ d} \qquad \text{but} \qquad 1 \text{ solar year} = 365.24 \text{ d} \qquad \Rightarrow \qquad \text{shortfall} \approx 10.9 \text{ d per year.}$$

Without a correction, festivals would drift through all the seasons. So the Hindu *lunisolar* calendar adds an extra month, the **Adhik Maas** (also called *Purushottam Maas*), about every **32½ months — 7 times in 19 years**.

Why 7 in 19? Because 19 solar years almost exactly equal 235 lunar months:

$$19 \times 365.2422 = 6939.60 \text{ d}, \qquad 235 \times 29.5306 = 6939.69 \text{ d}, \qquad 235 = 19 \times 12 + \mathbf{7}.$$

(In Greece this is called the *Metonic cycle*.)

The rule: a lunar month in which the Sun does **not** enter a new rashi (no Sankranti occurs) is an Adhik Maas. Recent examples: Adhik Shravana in **2023** (July–August) and Adhik Jyeshtha in **2026** (May–June).

6. Nakshatras: the 27 lunar mansions

The Moon takes 27.3 days to go round the zodiac, so ancient Indian astronomers divided its path into 27 equal parts:

$$\frac{360^\circ}{27} = 13^\circ 20' \quad \text{each — the Moon spends about one day in each nakshatra.}$$

Each nakshatra is marked by a bright "junction star" (*yogatara*):

#	Nakshatra	Main star(s)	#	Nakshatra	Main star(s)
1	Ashwini	β, γ Arietis	15	**Swati**	**Arcturus**
2	Bharani	35, 39, 41 Arietis	16	Vishakha	α, β Librae
3	**Krittika**	**Pleiades**	17	Anuradha	β, δ, π Scorpii
4	**Rohini**	**Aldebaran**	18	**Jyeshtha**	**Antares**
5	Mrigashira	λ Orionis	19	Mula	tail of Scorpius (Shaula)
6	**Ardra**	**Betelgeuse**	20	Purva Ashadha	δ, ε Sagittarii
7	**Punarvasu**	**Castor & Pollux**	21	Uttara Ashadha	σ, ζ Sagittarii
8	Pushya	γ, δ, θ Cancri	22	**Shravana**	**Altair**
9	Ashlesha	head of Hydra	23	Dhanishtha	Delphinus
10	**Magha**	**Regulus**	24	Shatabhisha	λ Aquarii
11	Purva Phalguni	δ, θ Leonis	25	Purva Bhadrapada	α, β Pegasi
12	Uttara Phalguni	Denebola	26	Uttara Bhadrapada	γ Pegasi, α Andromedae
13	Hasta	Corvus (the "hand")	27	Revati	ζ Piscium

(continued)

#	Nakshatra	Main star(s)	#	Nakshatra	Main star(s)
14	**Chitra**	**Spica**	(28)	(Abhijit)	(Vega, in older lists)



Five neighbouring nakshatras drawn with Stellarium's "Indian nakshatras" sky culture: Krittika (the Pleiades), Rohini (around Aldebaran), Mrigashira, Ardra (Betelgeuse) and Punarvasu (Castor & Pollux).

Memory tip: the bold ones contain bright stars you can spot even from a city — learn these first. The 12 **rashis** each span 30° ($= 2\frac{1}{4}$ nakshatras): Mesha, Vrishabha, Mithuna, Karka, Simha, Kanya, Tula, Vrischika, Dhanu, Makara, Kumbha, Meena.

Did you know?

The *Vedanga Jyotisha*, one of the oldest Indian astronomy texts (late 1st millennium BCE), already used the nakshatras and a 5-year lunisolar cycle to time rituals.

7. Indian units of time

The *Surya Siddhanta* defines a neat base-60 system:

Unit	Definition	Modern value
Prana (asu)	one breath (10 long syllables)	**4 s**
Pala (vinadi)	6 pranas	**24 s**
Ghati (ghatika, nadi)	60 palas	**24 min**
Muhurta	2 ghatis	**48 min**

(continued)

Unit	Definition	Modern value
Prahar (yama)	$\frac{1}{8}$ of a day	**3 h**
Ahoratra (day + night)	60 ghatis = 30 muhurtas	**24 h**

Worked examples

1. Ghatis in 6 hours: $\dfrac{6 \times 60 \text{ min}}{24 \text{ min}} = \mathbf{15}$ ghatis.

2. "Brahma Muhurta" traditionally begins 2 muhurtas before sunrise. If sunrise is at 6:20 AM: $2 \times 48 = 96$ min $= 1$ h 36 min, so it begins at **4:44 AM**.

The ghati was measured with a water clock (*ghatika yantra*): a copper bowl with a tiny hole that sinks in a tub of water in exactly one ghati.

8. Indian eras and the National Calendar

Era	Epoch	Conversion (from its New Year)
Vikram Samvat	57 BCE	VS = CE +57 (VS 2083 began in March 2026)
Shaka Samvat	78 CE	Shaka = CE −78 (Shaka 1948 began 22 March 2026)
Kali Yuga (traditional)	18 Feb 3102 BCE	Kali = CE +3101

Indian connection — the Indian National Calendar (1957)

After Independence, the **Calendar Reform Committee**, chaired by the astrophysicist **Meghnad Saha**, studied the more than 30 calendars then in use. Its recommendation — the **Indian National Calendar** (Saka calendar) — was adopted on **22 March 1957**. It is solar: **1 Chaitra = 22 March** (21 March in leap years); Chaitra has 30 days (31 in leap years), Vaishakha to Bhadra have 31, and Ashvin to Phalguna have 30. It is used alongside the Gregorian date in the Gazette of India.



The Vrihat Samrat Yantra at Jantar Mantar, Jaipur — a giant sundial built by Maharaja Sawai Jai Singh II in the 1730s. Its sloping wall points at the celestial pole, and the shadow it casts tells local time to about 2 seconds.

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Jantar Mantar

Between the 1720s and 1730s Sawai Jai Singh II built five stone observatories — Delhi, Jaipur, Ujjain, Varanasi and Mathura. The Samrat Yantra in Jaipur is about **27 m tall**; its gnomon is tilted at Jaipur's latitude ($\approx 27°$). Jaipur's Jantar Mantar is a UNESCO World Heritage Site.

Key points to remember

- Weekday order = Chaldean order (Saturn, Jupiter, Mars, **Sun**, Venus, Mercury, Moon), **jumping 3 places** each day, because $24 = 3 \times 7 + 3$.
- **Tithi =** $12°$ **of Moon–Sun angle**; 30 tithis per month; Shukla 1–15 (ends at Purnima), Krishna 16–30 (ends at Amavasya).
- Months are named after the **nakshatra near the Full Moon** (Chitra → Chaitra, Krittika → Kartika...).
- **Adhik Maas** $\approx$ every 32½ months (7 in 19 years) keeps festivals in their seasons.
- 27 nakshatras $\times\ 13°20' = 360°$; the Moon spends about a day in each.
- 1 ghati = 24 min; 1 muhurta = 48 min; 1 pala = 24 s.

Check yourself

1. Starting from Thursday (Jupiter), which planet rules the first hora of the next day? Check that it is Friday's planet.
2. The Moon is $175°$ ahead of the Sun. Which tithi is it?
3. Why does the Kartika Full Moon appear near the Pleiades?
4. Convert 3 ghatis 30 palas into minutes.

Answers: 1. 3 places on from Jupiter: Mars, Sun, **Venus** = Friday ✔. 2. $175/12 = 14.6 \Rightarrow$ 15th tithi, Purnima. 3. The month is named after Krittika because its Full Moon occurs there. 4. $3 \times 24 + 30 \times 0.4 = 84$ minutes.

Further reading (free / public)

- *Report of the Calendar Reform Committee* (Government of India, 1955; chair M. N. Saha) — freely readable on the Internet Archive (archive.org).
- OpenStax, *Astronomy 2e*, §4.3 "Keeping Time" and §4.4 "The Calendar" — `https://openstax.org/books/astronomy-2e/pages/4-4-the-calendar`
- S. Balachandra Rao, *Indian Astronomy: An Introduction* (Universities Press) — a readable book on the Panchang and Indian astronomy.
- **Stellarium**: switch the sky culture to "Indian Vedic" to see the nakshatras drawn on the sky.